

Homework 1

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Abstract

This is my abstract!

**HOMEWORK
SOLUTION**

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1. QUESTION 1

1.a. Proof

Start by taking logs on both sides of the return expression:

$$\log R_{e,t+1} = \log(1 + \exp(\log D_{t+1} - \log P_{t+1})) + \log P_{t+1} - \log P_t \quad (1)$$

For simplicity, I denote $r_{e,t+1} = \log R_{e,t+1}$, $p_{t+1} = \log P_{t+1}$, $d_{t+1} = \log D_{t+1}$, and $dp_{t+1} = \log D_{t+1} - \log P_{t+1}$, then, by a Taylor expansion in dp_{t+1} (which is stationary with mean $\bar{dp}$), we get

$$r_{e,t+1} = \log(1 + \exp(\bar{dp})) + \frac{\exp(\bar{dp})}{1 + \exp(\bar{dp})} (dp_{t+1} - \bar{dp}) + p_{t+1} - p_t. \quad (2)$$

Denote $\kappa = 1/(1 + \exp(\bar{dp}))$, and $\kappa_0 = \log(1 + \exp(\bar{dp})) - \bar{dp} \exp(\bar{dp})/(1 + \exp(\bar{dp})) = -\log \kappa - (1 - \kappa) \log(1/\kappa - 1)$. The log-linearization becomes

$$r_{e,t+1} = \kappa_0 + (1 - \kappa)dp_{t+1} + p_{t+1} - p_t. \quad (3)$$

Moving p_t to the left-hand side and the other terms to the right-hand side, we get

$$\begin{aligned} p_t &= \kappa_0 + (1 - \kappa)d_{t+1} - r_{e,t+1} + \kappa p_{t+1} \\ &= \kappa_0 + (1 - \kappa)d_{t+1} - r_{e,t+1} + \kappa [\kappa_0 + (1 - \kappa)d_{t+2} - r_{e,t+2} + \kappa p_{t+2}] \\ &= \kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\ &\quad + (1 - \kappa)(d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1}d_{t+H}) \\ &\quad - [r_{e,t+1} + \kappa r_{e,t+2} + \dots + \kappa^{H-1}r_{e,t+H}] \\ &\quad + \kappa^H p_{t+H} \end{aligned} \quad (4)$$

The log dividend-price ratio dp_t can then be written as

$$\begin{aligned} dp_t &= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\ &\quad + [r_{e,t+1} + \kappa r_{e,t+2} + \dots + \kappa^{H-1}r_{e,t+H}] \\ &\quad - d_t + (1 - \kappa)(d_{t+1} + \kappa d_{t+2} + \dots + \kappa^{H-1}d_{t+H}) \\ &\quad - \kappa^H p_{t+H} \\ &= -\kappa_0(1 + \kappa + \dots + \kappa^{H-1}) \\ &\quad + [r_{e,t+1} + \kappa r_{e,t+2} + \dots + \kappa^{H-1}r_{e,t+H}] \\ &\quad - d_t + d_{t+1} + \kappa(d_{t+1} - d_{t+2}) + \dots + \kappa^{H-1}(d_{t+H-1} - d_{t+H}) \\ &\quad + \kappa^H d_{t+H} - \kappa^H p_{t+H} \\ &= \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H dp_{t+H} \end{aligned} \quad (5)$$

where $\Delta d_{t+1} := d_{t+1} - d_t$. Notice that the identity above holds ex post (state by state), so it also holds in conditional expectation at time t , giving

$$dp_t = \kappa_0 \frac{\kappa^H - 1}{1 - \kappa} + \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} + \kappa^H \mathbb{E}_t dp_{t+H} \quad (6)$$

Letting $H \rightarrow \infty$ and imposing the no-bubble (transversality) condition

$$\lim_{H \rightarrow \infty} \kappa^H \mathbb{E}_t dp_{t+H} = 0 \quad (7)$$

we obtain equation (1.3):

$$dp_t = \kappa_0 \frac{-\kappa_0}{1 - \kappa} + \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^{\infty} \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \quad (8)$$

1.b.

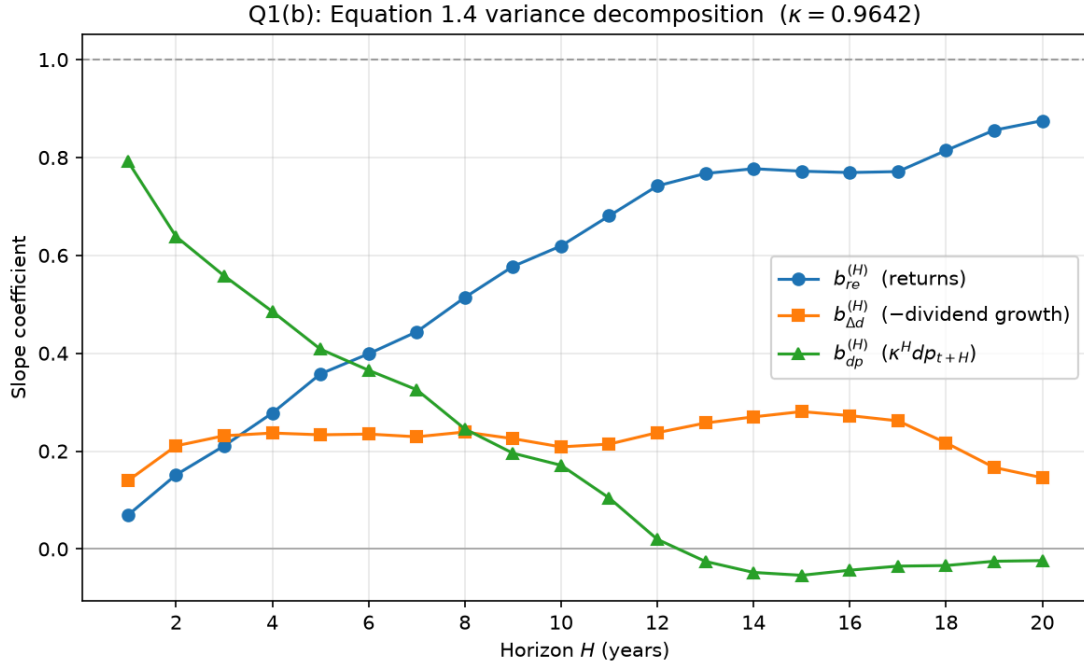


Figure 1: Equation 1.4 variance-decomposition slopes vs. horizon H (EQ Dataset, 1928–2021; overlapping monthly starts; $\kappa = 1/(1 + e^{\bar{dp}}) = 0.9642$).

We can infer from the figure that, valuation ratio dp_t can predict log run return, and the volatility of price mainly comes from the volatility of discount rate.

1.c. VAR

Using equation

$$b_{re}^{(H)} = 1_{re}(\Gamma + \kappa^H \Gamma^{H+1})(I - \kappa \Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (9)$$

$$b_{\Delta d}^{(H)} = -1_{\Delta d}(\Gamma + \kappa^H \Gamma^{H+1})(I - \kappa \Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (10)$$

and $b_{dp}^{(H)} = 1 - b_{re}^{(H)} - b_{\Delta d}^{(H)}$.

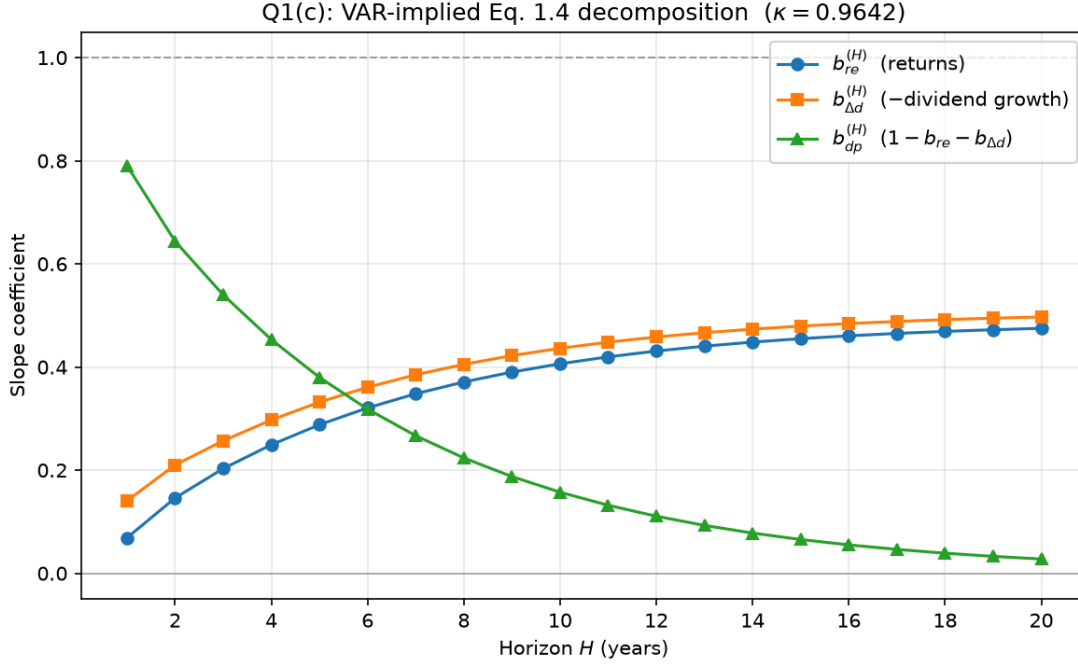


Figure 2: VAR-implied Equation 1.4 variance-decomposition terms vs. horizon H . Overlapping-annual VAR(1) on $Z_t = [\Delta d_t, r_{e,t}, dp_t]'$ estimated by OLS (EQ Dataset, 1928–2021; $\kappa = 0.9642$, as in 1(b)). $b_{dp}^{(H)}$ is imposed as $1 - b_{re}^{(H)} - b_{\Delta d}^{(H)}$.

1.d.

As $H \rightarrow \infty$, the

$$b_{re}^{(\infty)} = 1_{re} \Gamma(I - \kappa \Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (11)$$

$$b_{\Delta d}^{(\infty)} = -1_{\Delta d} \Gamma(I - \kappa \Gamma)^{-1} \frac{Cov(z_t, dp_t)}{Var(dp_t)} \quad (12)$$

The calculated numbers are $b_{re}^{(\infty)} = 0.4892$, $b_{\Delta d}^{(\infty)} = 0.5096$, and $b_{dp}^{(\infty)} = 0.0012$. These numbers show that valuation ratio dp can both predict dividend growth and returns, and half of price volatility is attribute to dividend growth variance, half is attribute to discount rate variance. This result is not aligned with Cochrane(2011).

1.e. Alternative log-linear present-value identity

Using the same trick,

$$R_{e,t+1} = \frac{P_{t+1} + D_{t+1}}{D_{t+1}} \frac{D_{t+1}}{D_t} \frac{D_t}{P_t + D_t} \frac{P_t + D_t}{P_t} \quad (13)$$

, the log one-period return can be written as

$$r_{e,t+1} = -\log(1 - e^{-dy_{t+1}}) + \Delta d_{t+1} + \log(1 - e^{-dy_t}) + dy_t \quad (14)$$

where $dy_t := \log\left(1 + \frac{D_t}{P_t}\right)$ and $\log(1 - e^{-dy_t}) = \log \frac{D_t}{P_t + D_t}$.

Applying a Taylor expansion of $\log(1 - e^{-dy_t})$ in dy_t around its mean $\bar{dy}$, we have

$$\log(1 - e^{-dy_t}) = \log(1 - e^{-\bar{d}y}) + \frac{e^{-\bar{d}y}}{1 - e^{-\bar{d}y}}(dy_t - \bar{d}y) + O((dy_t - \bar{d}y)^2) \quad (15)$$

Letting $\kappa = e^{-\bar{d}y}$ to simplify the expression,

$$\log(1 - e^{-dy_t}) = \log(1 - \kappa) + \frac{\kappa}{1 - \kappa}(dy_t - \bar{d}y) + O((dy_t - \bar{d}y)^2) \quad (16)$$

Then the return is log-linearized as

$$r_{e,t+1} = -\frac{\kappa}{1 - \kappa}(dy_{t+1} - \bar{d}y) + \Delta d_{t+1} + \frac{\kappa}{1 - \kappa}(dy_t - \bar{d}y) + dy_t \quad (17)$$

and

$$\begin{aligned} \frac{dy_t}{1 - \kappa} &= r_{e,t+1} - \Delta d_{t+1} + \kappa \frac{dy_{t+1}}{1 - \kappa} \\ &= r_{e,t+1} - \Delta d_{t+1} \\ &\quad + \kappa \left[r_{e,t+2} - \Delta d_{t+2} + \kappa \frac{dy_{t+2}}{1 - \kappa} \right] \\ &= r_{e,t+1} - \Delta d_{t+1} + \kappa [r_{e,t+2} - \Delta d_{t+2}] \\ &\quad + \dots + \kappa^{H-1} [r_{e,t+H} - \Delta d_{t+H}] + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \\ &= \sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} + \kappa^H \frac{dy_{t+H}}{1 - \kappa} \end{aligned} \quad (18)$$

Thus, the log-linearization identity is

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \Delta d_{t+h} \right) + \kappa^H dy_{t+H} \quad (19)$$

Taking the conditional expectation at time t , the identity also holds

$$dy_t = (1 - \kappa) \left(\sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t r_{e,t+h} - \sum_{h=1}^H \kappa^{h-1} \mathbb{E}_t \Delta d_{t+h} \right) + \kappa^H \mathbb{E}_t dy_{t+H} \quad (20)$$

Imposing the no-bubble assumption, the last term vanishes as $h \rightarrow \infty$,

$$\lim_{H \rightarrow \infty} \kappa^H E_t \log \left(\frac{P_t + D_t}{P_t} \right) \rightarrow 0 \quad (21)$$

2. QUESTION 2

2.a.

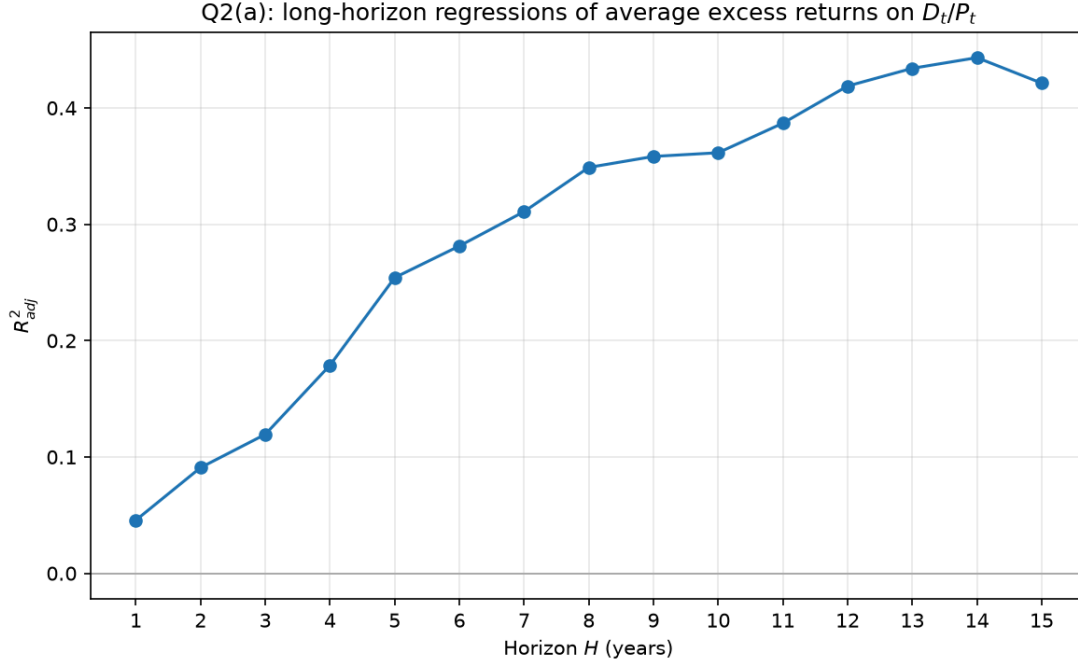


Figure 3: Adjusted R^2 of the regressions of the average future excess equity return $\frac{1}{H} \sum_{h=1}^H x R_{e,t+h}$ on D_t/P_t , plotted against the horizon $H = 1, \dots, 15$ years.

Valuation ratio $\frac{D}{P}$ predicts returns in the long run better than in the short run.

2.b.

Estimating $xR_{e,t+1} = a + b \cdot D_t/P_t + \varepsilon_t$ by OLS on $T = 1117$ monthly start dates (1927:12–2020:12) gives $\hat{a} = -0.0314$ and $\hat{b} = 2.8038$. The five standard errors for $\hat{b}$, all built from $\widehat{\text{Var}}[\hat{\theta}] = \frac{1}{T} Q^{-1} \hat{S} Q^{-1}$ with $Q = \frac{1}{T} \sum_{\tau} x_{\tau} x_{\tau}'$, are:

Standard error method	$\hat{b}$	s.e. $(\hat{b})$	t -statistic
(i) OLS	2.8038	0.3798	7.38
(ii) White (1980)	2.8038	0.6587	4.26
(iii) Newey-West (1987), 11 lags	2.8038	1.2990	2.16
(iv) Hansen-Hodrick (1980), 11 lags	2.8038	1.4537	1.93
(v) Newey-West (1987, 1994), $L = 24$	2.8038	1.2304	2.28

The lag length in (v) is data-driven: the Newey and West (1994, Eq. 2.2) rule gives $L = a \cdot T^{1/3}$ with $a = 2.31$, hence $L = 24$. The Hansen-Hodrick estimate in (iv) is positive definite in this sample, so no truncation was needed.

2.c.

I report Amihud and Hurvich (2004) estimation.

$$\frac{D_{t+1}}{P_{t+1}} = \frac{\hat{\theta}}{0.0111} + \frac{\hat{\phi}}{0.7197} \frac{D_t}{P_t} + \hat{\varepsilon}_{t+1} \quad (22)$$

$$\hat{\phi}^c = \hat{\phi} + \frac{1}{T}(1 + 3\hat{\phi}) + \frac{3}{T^2}(1 + 3\hat{\phi}) = 0.7540, \quad T = 95 \quad (23)$$

$$\hat{u}_{t+1}^c = \frac{D_{t+1}}{P_{t+1}} - \left(\hat{\theta} + \hat{\phi}^c \frac{D_t}{P_t} \right) \quad (24)$$

$$xR_{e,t+1} = \underset{-0.0314}{\hat{a}} + \underset{2.3412}{\hat{b}} \frac{D_t}{P_t} + \underset{-13.4837}{\hat{b}_u} \hat{u}_{t+1}^c + \hat{\varepsilon}_{t+1} \quad (25)$$

Amihud and Hurvich estimation gives a smaller value than OLS estimation. This is because OLS estimation is upper biased in small sample as innovations of dividend-price ratio and excess return are positive correlated, and the AR(1) estimation is lower biased.

2.d.

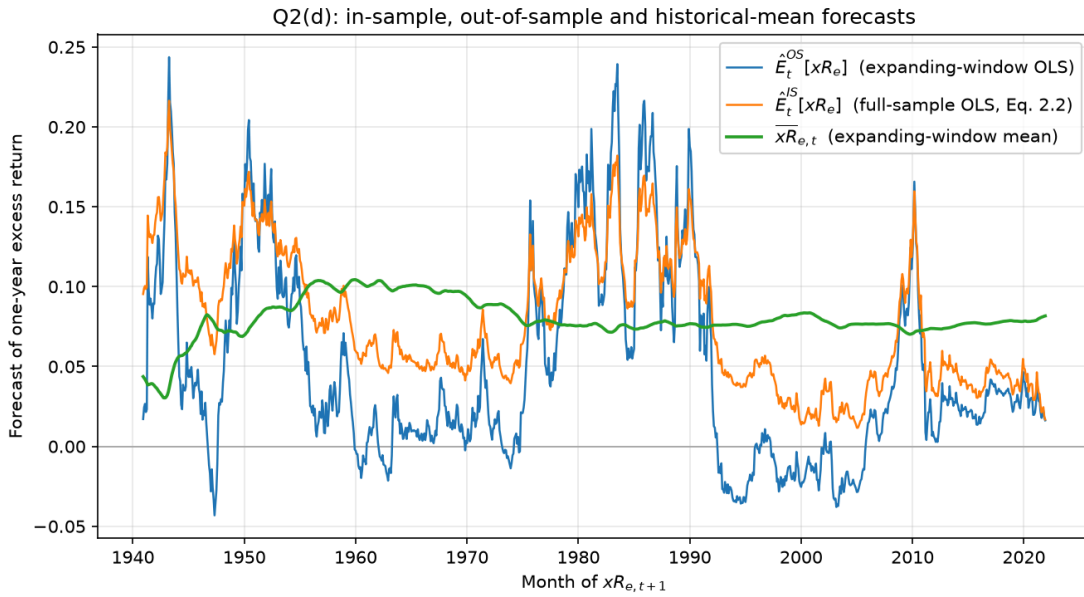


Figure 4: Forecasts of the one-year-ahead excess return for $t + 1$ from December 1940 to December 2021: the out-of-sample forecast $\hat{E}_t^{OS}[xR_e] = a_t + b_t D_t/P_t$ with a_t, b_t estimated on an expanding window, the in-sample forecast $\hat{E}_t^{IS}[xR_e] = \hat{a} + \hat{b} D_t/P_t$ from Equation 2.2, and the expanding-window historical mean $\overline{xR}_{e,t}$.

Each expanding window runs from the pair $(t, t + 1) = (\text{December } 1927, \text{December } 1928)$ through the pair one month before the forecast pair (for the first forecast, through (November 1939, November 1940), 144 pairs), and $\overline{xR}_{e,t}$ averages $xR_{e,t+1}$ over the same window. Over the 973 forecasts with $t + 1$ from December 1940 to December 2021,

$$R_{OS}^2 = 1 - \frac{\sum_t (xR_{e,t+1} - \hat{E}_t^{OS}[xR_e])^2}{\sum_t (xR_{e,t+1} - \mu_{xR})^2} = 1 - \frac{26.7838}{26.8327} = 0.0018 \quad (26)$$

where μ_{xR} is the sample mean of $xR_{e,t+1}$ over the same 973 months; no degrees-of-freedom adjustment is applied.

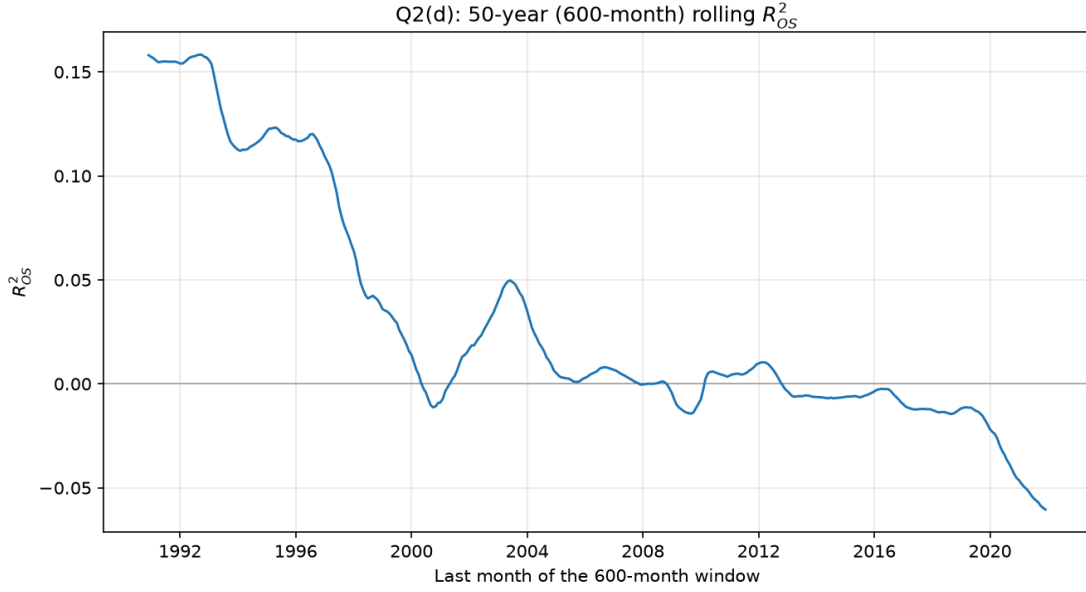


Figure 5: R^2_{OS} over 600-month (50-year) rolling windows of the out-of-sample forecast errors, plotted at each window's last month from December 1990 to December 2021 (first window: January 1941 to December 1990). a_t and b_t are the expanding-window estimates above; SST uses the sample mean of $xR_{e,t+1}$ within each window.

2.e.

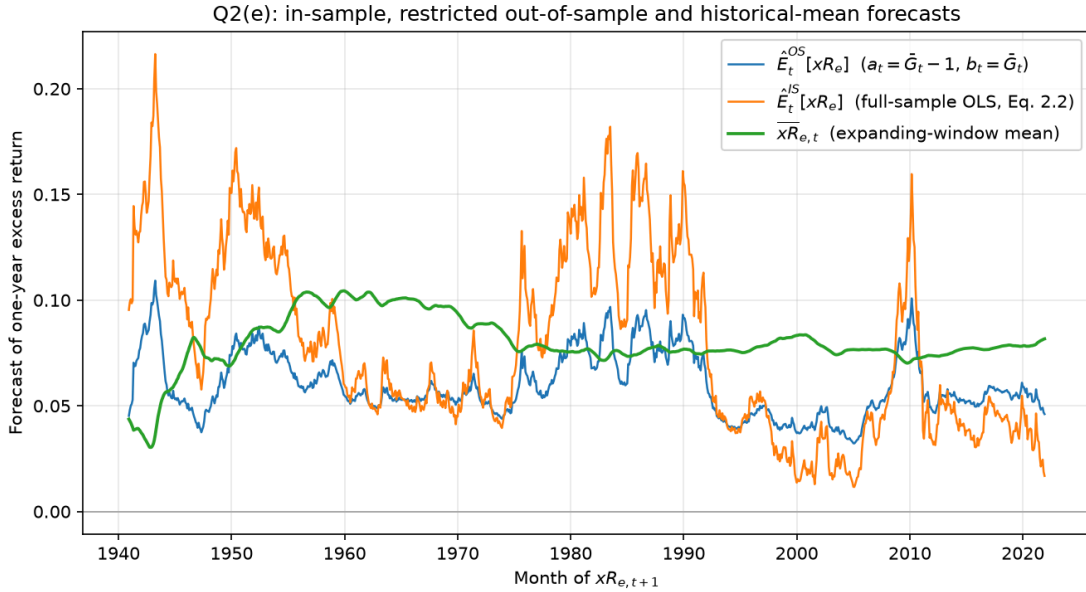


Figure 6: Forecasts of the one-year-ahead excess return for $t + 1$ from December 1940 to December 2021: the restricted out-of-sample forecast $\hat{E}_t^{OS}[xR_e] = (\bar{G}_t - 1) + \bar{G}_t D_t/P_t$, the in-sample forecast $\hat{E}_t^{IS}[xR_e] = \hat{a} + \hat{b} D_t/P_t$ from Equation 2.2, and the expanding-window historical mean $\bar{xR}_{e,t}$.

The sample, forecast dates, expanding windows, $\hat{E}_t^{IS}[xR_e]$ and $\bar{xR}_{e,t}$ are those of Question 2(d). $\bar{G}_t$ averages $e^{\Delta d}$ over the same expanding window; for the first forecast it uses Δd from December 1927 to

November 1939 (144 months), so $\bar{G}_t = 1.0000$. Over the 973 forecasts with $t + 1$ from December 1940 to December 2021,

$$R_{OS}^2 = 1 - \frac{\sum_t (xR_{e,t+1} - \hat{E}_t^{OS}[xR_e])^2}{\sum_t (xR_{e,t+1} - \mu_{xR})^2} = 1 - \frac{26.7797}{26.8327} = 0.0020 \quad (27)$$

where, as in Question 2(d), μ_{xR} is the sample mean of $xR_{e,t+1}$ over the same 973 months and no degrees-of-freedom adjustment is applied.

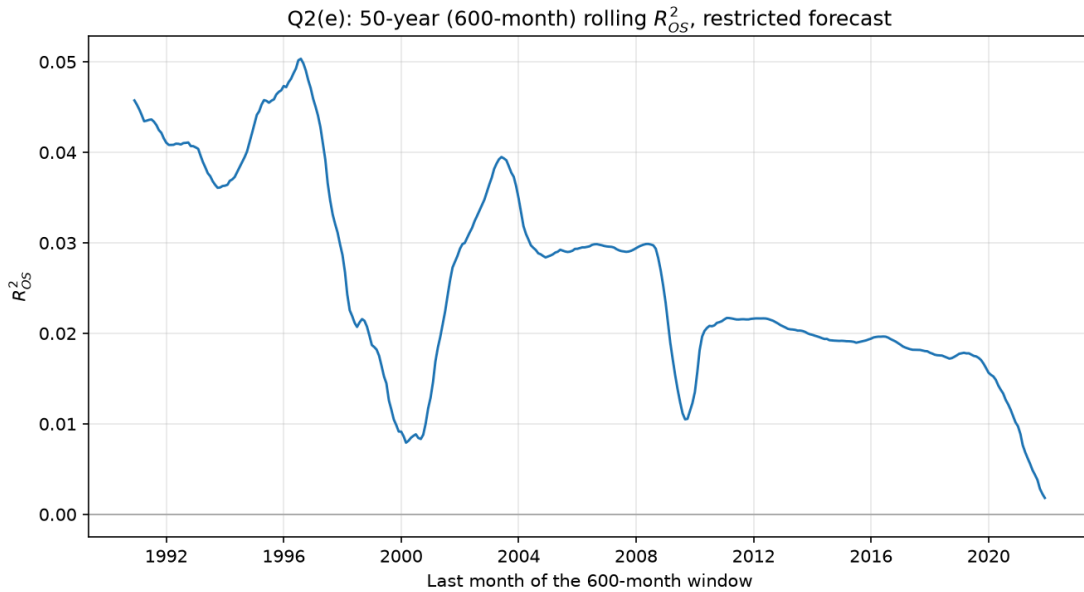


Figure 7: R_{OS}^2 of the restricted forecast over 600-month (50-year) rolling windows of the out-of-sample forecast errors, plotted at each window's last month from December 1990 to December 2021 (first window: January 1941 to December 1990). SST uses the sample mean of $xR_{e,t+1}$ within each window.

3. QUESTION 3

3.a. 3(a)

For each month τ I estimate the cross-firm regression

$$\text{MOM}_{j,\tau}^{CZ} = a_\tau + b_\tau \cdot \text{MOM}_{j,\tau} + \varepsilon_{j,\tau} \quad (28)$$

and plot the resulting $\hat{a}_\tau$, $\hat{b}_\tau$ and R_τ^2 below. Two versions of my own signal are shown: one compounding the 11 returns over $\tau - 12, \dots, \tau - 2$ (skipping the most recent month) and one compounding all 12 returns over $\tau - 12, \dots, \tau - 1$, which is the window used in the problem statement and by Chen and Zimmermann (2022).

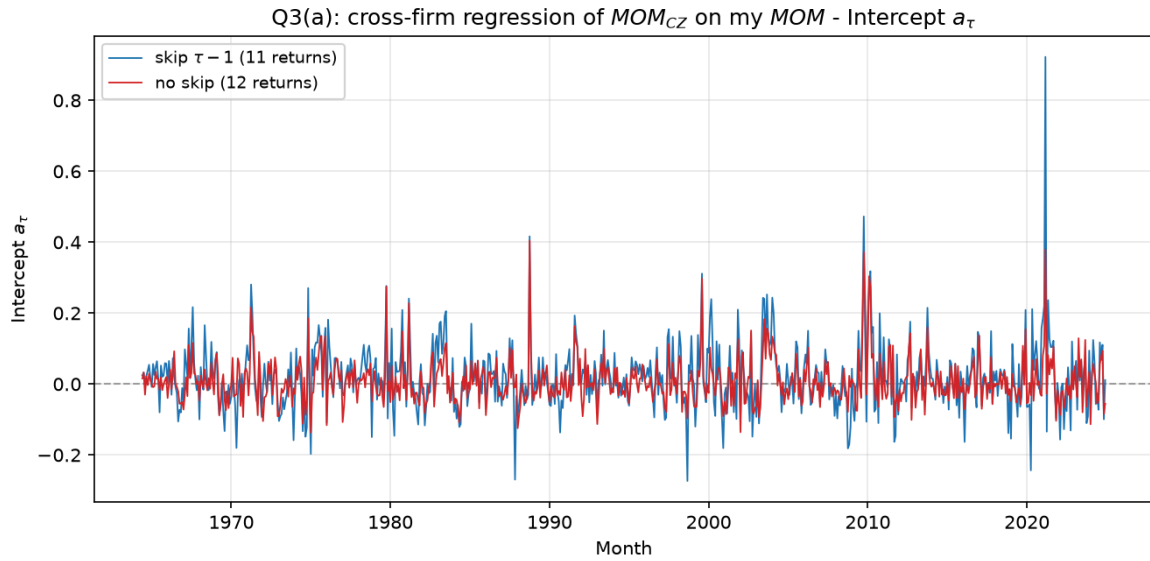


Figure 8: Monthly intercepts $\hat{a}_\tau$ from the cross-firm regression of MOM_{CZ} on my MOM , June 1964 – December 2024 (727 months, on average 3,351 firms per month).

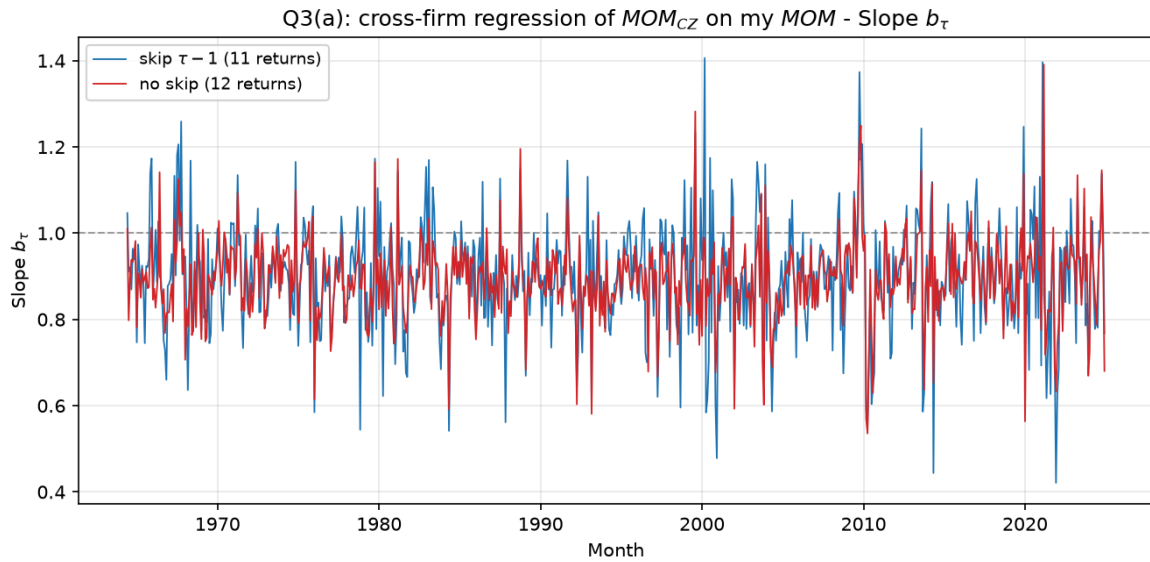
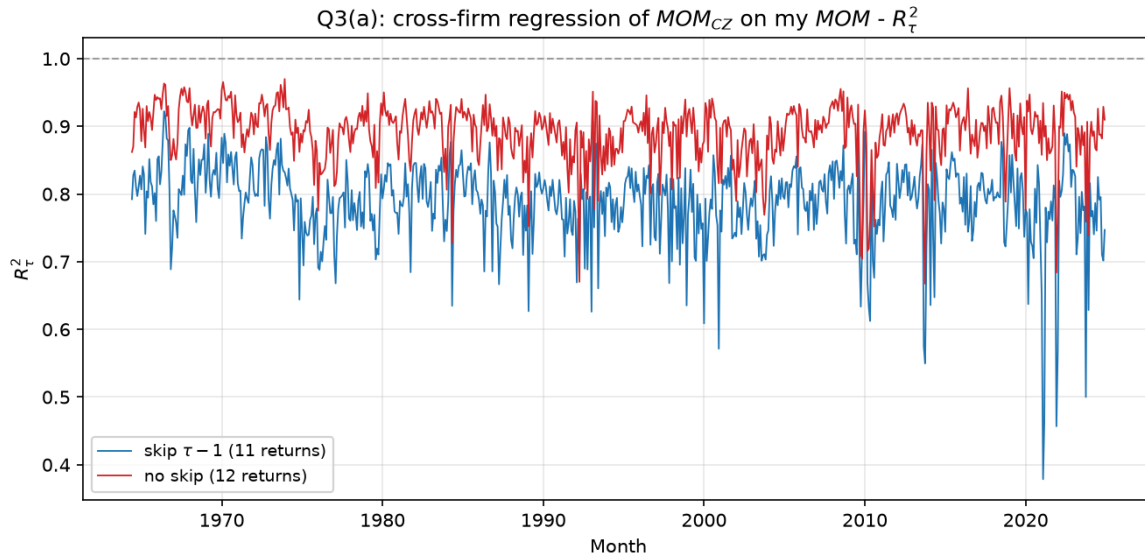


Figure 9: Monthly slopes $\hat{b}_\tau$.

Figure 10: Monthly R^2_τ .

4. QUESTION 4

4.a.

Average excess log yields, log forward rates and log annual returns (Fama-Bliss discount bonds; xy averaged over 871 months from June 1952 to December 2024, xf over 871 months from June 1952 to December 2024, and xr over 859 months from June 1953 to December 2024), in percent:

H	average $xy_{b,t}^{(H)}$	average $xf_{b,t}^{(H)}$	average $xr_{b,t}^{(H)}$
2	0.1686	0.3372	0.3154
3	0.3278	0.6461	0.6069
4	0.4660	0.8805	0.8198
5	0.5639	0.9557	0.8719

where $xy_{b,t}^{(H)} = y_{b,t}^{(H)} - y_{b,t}^{(1)}$, $xf_{b,t}^{(H)} = f_{b,t}^{(H)} - y_{b,t}^{(1)}$ and $xr_{b,t}^{(H)} = r_{b,t}^{(H)} - r_{b,t}^{(1)}$, with $y_{b,t}^{(H)} = \log(1 + Y_{b,t}^{(H)})$, $f_{b,t}^{(H)} = H y_{b,t}^{(H)} - (H-1) y_{b,t}^{(H-1)}$ and $r_{b,t}^{(H)} = H y_{b,t-1}^{(H)} - (H-1) y_{b,t}^{(H-1)}$, where $t-1$ is the same month one year earlier.

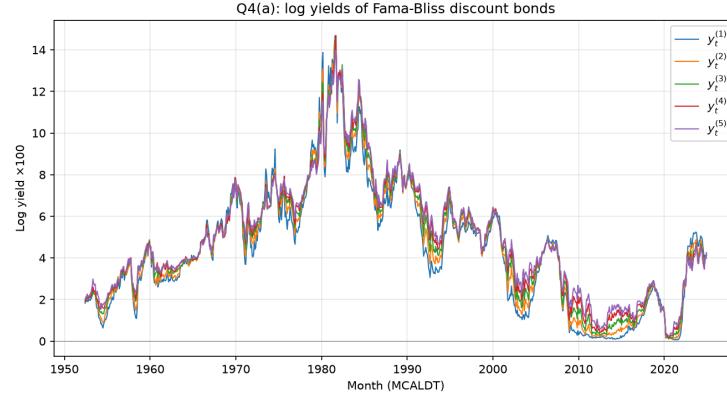


Figure 11: Log yields $y_{b,t}^{(H)}$ of the Fama-Bliss discount bonds for $H = 1, \dots, 5$, June 1952 to December 2024 (shown $\times 100$).

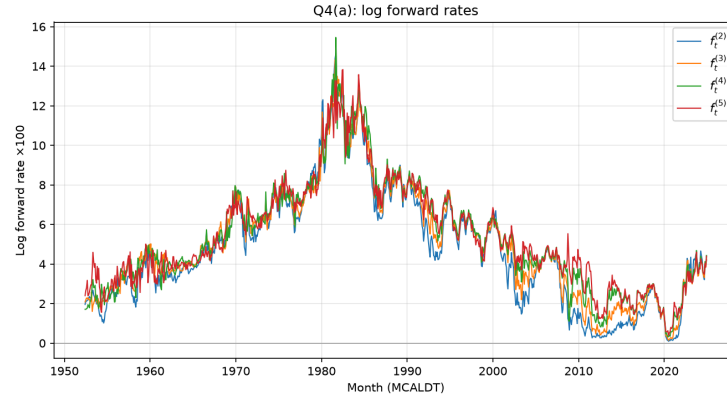


Figure 12: Log forward rates $f_{b,t}^{(H)}$ for $H = 2, \dots, 5$, June 1952 to December 2024 (shown $\times 100$).

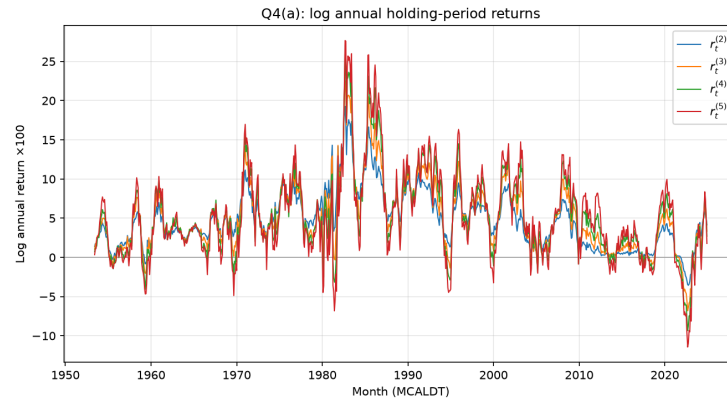


Figure 13: Log annual returns $r_{b,t}^{(H)}$ for $H = 2, \dots, 5$, June 1953 to December 2024 (shown $\times 100$).

4.b.

Regressions of the average annual hold-to-maturity excess return on the excess log yield,

$$\frac{1}{H} xr_{b,t:t+H}^{(H)} = a^{(H)} + b^{(H)} xy_{b,t}^{(H)} + \varepsilon_t^{(H)}, \quad xr_{b,t:t+H}^{(H)} = \sum_{h=1}^H xr_{b,t+h}^{(H-h+1)}, \quad (29)$$

estimated by OLS on overlapping monthly observations, where $t + h$ is the same month h years later and $xr_{b,t}^{(1)} = 0$. The t -statistics use Hansen and Hodrick (1980) standard errors with $L = 12H - 1$ monthly lags, the number of months by which consecutive H -year dependent variables overlap; each regression uses every month t for which its variables are available.

H	$b^{(H)}$	t -statistic (Hansen-Hodrick)
2	0.6774	3.57
3	0.5316	3.21
4	0.4141	2.52
5	0.3453	2.30

4.c.

Regressions of the one-year excess log return on the excess log forward rate,

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} x f_{b,t}^{(H)} + \varepsilon_{t+1}^{(H)}, \quad (30)$$

estimated by OLS on overlapping monthly observations, where $t + 1$ is the same month one year later, using the 859 months with t from June 1952 to December 2023 for every H . The t -statistics use Newey and West (1987, 1994) standard errors: Bartlett weights with the lag length chosen by the Newey and West (1994) rule, as in Question 2(b) ($L = 23, 23, 24, 23$ for $H = 2, 3, 4, 5$).

H	$b^{(H)}$	t -statistic (Newey-West)
2	0.6774	3.23
3	0.8887	3.35
4	1.1163	3.62
5	0.9641	2.91

4.d.

Cochrane and Piazzesi (2005) factor, estimated from

$$\frac{1}{4} \sum_{H=2}^5 xr_{b,t+1}^{(H)} = \theta_0 + cp_t + u_t, \quad cp_t = \sum_{H=1}^5 \theta_H f_{b,t}^{(H)}, \quad (31)$$

by OLS on overlapping monthly observations, where $t + 1$ is the same month one year later and $f_{b,t}^{(1)} = y_{b,t}^{(1)}$, using the 859 months with t from June 1952 to December 2023.

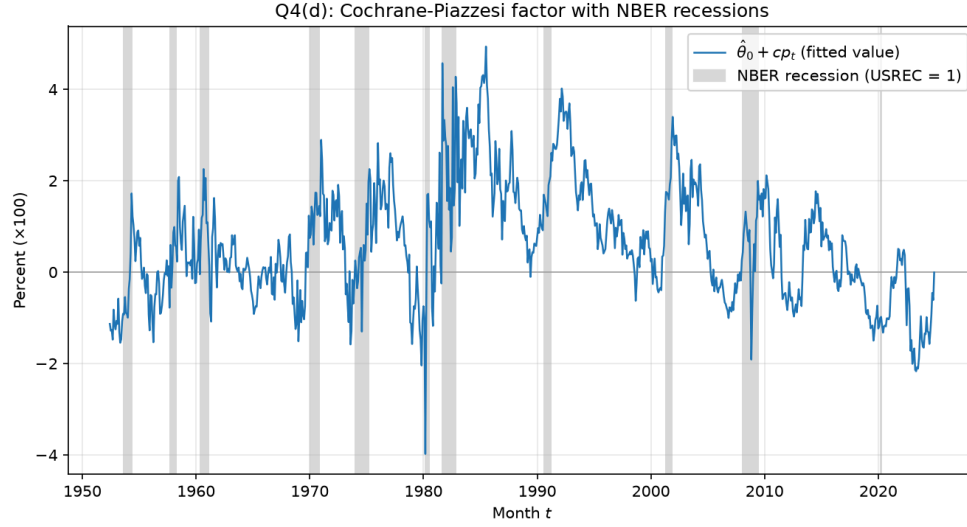


Figure 14: The figure plots the fitted value $\hat{\theta}_0 + cp_t$ of $\frac{1}{4} \sum_{H=2}^5 xr_{b,t+1}^{(H)}$ (shown $\times 100$) for t from June 1952 to December 2024; grey areas mark NBER recession months (FRED USREC = 1). Months after December 2023 use the coefficients estimated through December 2023.

4.e.

Regressions of the one-year excess log return on the Cochrane-Piazzesi factor,

$$xr_{b,t+1}^{(H)} = a^{(H)} + b^{(H)} (\hat{\theta}_0 + cp_t) + \varepsilon_{t+1}^{(H)}, \quad (32)$$

estimated by OLS on overlapping monthly observations, where $t + 1$ is the same month one year later, using the 859 months with t from June 1952 to December 2023 for every H . The regressor is the fitted value $\hat{\theta}_0 + cp_t$ plotted in Question 4(d). The t -statistics use Newey and West (1987, 1994) standard errors: Bartlett weights with the lag length chosen by the Newey and West (1994) rule, as in Question 2(b) ($L = 23, 23, 23, 23$ for $H = 2, 3, 4, 5$).

H	$b^{(H)}$	t -statistic (Newey-West)
2	0.4418	4.16
3	0.8274	4.18
4	1.2517	4.45
5	1.4791	4.25